| Question | | | | Marking details | Marks Available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| **2** | (a) | | | **Any three (x1) from:**<br>Temperature<br>Time<br>Rate of spin<br>Volume/ mass/ amount of DNA/ sample<br>relevant condition of tube ie {density / concentration / volume / height of/ type of } medium Not solution unqualified<br>3 correct = 2 marks, 2/1 correct = 1 mark, 0 correct = 0 marks | | 2 | | 2 | | 2 |
| | (b) | (i) | | A: 14N /light/14<br>B: $^{14}N + ^{15}N$ /light and heavy/14 and 15<br>C: $^{15}N$ / heavy/ 15<br>All correct = 1 mark | | 1 | | 1 | | 1 |
| | | (ii) | | A. {Generation 0 /{band/peak} at C} is all {heavy isotope/ $N^{15}$} (1)<br>B. **Generation 1** {band/ peak} at B is {a mixture of heavy and light isotope/ an intermediate}(1)<br>C. Because each molecule retains one strand of heavy isotope and one strand of newly formed DNA of light isotope (1)<br>Accept hybrid DNA for intermediate DNA | 2 | 1 | | 3 | | |
| | | (iii) | | A. (Only) {$^{14}N$ / light isotope} (nucleotides ) are available for replication (1)<br>B. **Generation 2** has equal {height scans/ peaks / density} (1)<br>C. **because** one strand of heavy and one strand of light DNA have been used as templates for the formation of new DNA molecules/ owtte (1)<br>D. **Generation 3** has a higher peak of {14N / light isotope/ A} **because** more light DNA (strands) are used as template/ owtte (1) | | 3 | 1 | 4 | | |

| Question | | | | Marking details | Marks Available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| | | (iv) | | **Any two for 1 mark from:**<br>DNA polymerase / (DNA) helicase / (DNA) ligase (1)<br>**Any two (x1) from:**<br>• DNA polymerase – join free nucleotides (to the ends of the new DNA strand)(1)<br>• (DNA) helicase – {unzip/ unwind} {DNA/ helix} / breaks H-bonds {between nucleotides/ within the double strand/ between the bases} / separate (DNA) strands (1)<br>• (DNA) ligase – joins fragment of DNA to an existing fragment(1) | 3 | | | 3 | | |
| | | | | **Question 2 total** | **5** | **7** | **1** | **13** | **0** | **3** |